Andrew Y.K. Foong

PERSONAL WEBSITE: https://andrewfoongyk.github.io/
GOOGLE SCHOLAR: scholar.google.com/citations?user=2UOjgIUAAAAJ
CONTACT: foong.andrew@mayo.edu

Machine learning researcher in protein generative modeling, clinical AI, and Bayesian inference. Co-first author of BioEmu (*Science*, 2025). Leading a clinical AI research team at the Mayo Clinic; working with physicians to develop, implement, and validate models.

EXPERIENCE

MAYO CLINIC — January 2025 – Present

AI Scientist, Senior Associate Consultant
Assistant Professor of Biochemistry & Molecular Biology · Assistant Professor of Biomedical Informatics

Co-directing the AI & Data Analytics (AIDA) group in Radiation Oncology. Working with clinicians and operations teams, and supervising post-doctoral researchers in defining research questions, identifying data and workflow constraints, training models, and deploying AI into high-volume cancer treatment workflows.

Research areas include agent scaffolding for clinical record analysis, machine learning for medical imaging, and predictive modeling of treatment outcomes.

MICROSOFT RESEARCH CAMBRIDGE — November 2022 – November 2024

Senior Researcher

Developed probabilistic generative models for protein dynamics in the AI for Science team, including mathematical formulation, implementation, and large-scale training runs. Co-first author of BioEmu (*Science*, 2025) and Timewarp (*NeurIPS*, 2023, spotlight). Contributed to SE(3) flow matching for protein backbone generation.

GOOGLE DEEPMIND — February 2022 – May 2022

Research Scientist Intern

Data-efficient and Bayesian learning under Dr. Michalis Titsias. Developed deep generative models that combine variational autoencoders with Gaussian-process structure.

MICROSOFT RESEARCH CAMBRIDGE — July 2021 – October 2021

Research Intern

Accelerating molecular simulation with deep learning under Dr. Sebastian Nowozin.

TEACHING AND RESEARCH SERVICE

Created AIHC 5100, *AI Fundamentals for Healthcare Professionals*, a 24-lecture, two-credit course for Mayo Clinic's AI in Healthcare master's program (2025). Supervised Cambridge University's Information Theory and Bayesian Inference courses (2018–2021).

Organizer, NeurIPS 2021 Approximate Inference in Bayesian Deep Learning Competition. Reviewer for JMLR, NeurIPS, and AISTATS.

EDUCATION

PH.D. IN ENGINEERING, UNIVERSITY OF CAMBRIDGE October 2018 – November 2022

Machine learning; supervisor: Professor Richard E. Turner.

Thesis: *Approximate Inference in Bayesian Neural Networks and Translation Equivariant Neural Processes.*

M.ENG. & B.A. IN ENGINEERING, UNIVERSITY OF CAMBRIDGE October 2014 – June 2018

First Class Honors with Distinction

Specialized in information and computer engineering. Top 1–2% of the year group in the first three years; Institution of Engineering and Technology Prize, 2018.

SELECTED RESEARCH PAPERS

Asterisks denote co-first authorship. Full publication list: Google Scholar.

PROTEIN GENERATIVE MODELING

1. Sarah Lewis*, Tim Hempel*, José Jiménez-Luna*, Michael Gastegger*, Yu Xie*, **ANDREW Y.K. FOONG***, ..., Frank Noé. *Scalable Emulation of Protein Equilibrium Ensembles with Generative Deep Learning*. Science, 2025.
2. Leon Klein*, **ANDREW Y.K. FOONG***, Tor Erlend Fjelde*, Bruno Młodozeniec*, ..., Frank Noé, Ryota Tomioka. *Timewarp: Transferable Acceleration of Molecular Dynamics by Learning Time-Coarsened Dynamics*. NeurIPS, 2023. (Spotlight.)
3. Jason Yim, Andrew Campbell, **ANDREW Y.K. FOONG**, ..., Regina Barzilay, Tommi S. Jaakkola, Frank Noé. *Fast Protein Backbone Generation with SE(3) Flow Matching*. NeurIPS Machine Learning for Structural Biology Workshop, 2023.
4. Jason Yim, Andrew Campbell, Emile Mathieu, **ANDREW Y.K. FOONG**, ..., Frank Noé, Regina Barzilay, Tommi S. Jaakkola. *Improved Motif-Scaffolding with SE(3) Flow Matching*. Transactions on Machine Learning Research, 2024.

PROBABILISTIC MACHINE LEARNING

5. **ANDREW Y.K. FOONG***, Wessel P. Bruinsma*, Jonathan Gordon*, ..., Richard E. Turner. *Meta-Learning Stationary Stochastic Process Prediction with Convolutional Neural Processes*. NeurIPS, 2020.
6. **ANDREW Y.K. FOONG***, David R. Burt*, Yingzhen Li, Richard E. Turner. *On the Expressiveness of Approximate Inference in Bayesian Neural Networks*. NeurIPS, 2020.
7. **ANDREW Y.K. FOONG***, Wessel P. Bruinsma*, David R. Burt, Richard E. Turner. *How Tight Can PAC-Bayes be in the Small Data Regime?* NeurIPS, 2021.
8. Jonathan Gordon*, Wessel P. Bruinsma*, **ANDREW Y.K. FOONG**, James Requeima, Yann Dubois, Richard E. Turner. *Convolutional Conditional Neural Processes*. ICLR, 2020. (Oral presentation.)

CLINICAL AI

9. Shashank Yadav, David M. Routman, **ANDREW Y.K. FOONG**. *The Cost of Binarizing Survival Outcomes in Clinical Prognostic Modeling*. Machine Learning for Healthcare, 2026.
10. Shashank Yadav, David M. Routman, **ANDREW Y.K. FOONG**. *Probing Clinical Concepts in an EHR Foundation Model via Sparse Autoencoders*. ICML Workshop on Mechanistic Interpretability, 2026.
11. Jason Holmes, Federico Mastroleo, Mariana Borras-Osorio, ..., **ANDREW Y.K. FOONG**, Wei Liu, Mark R. Waddle. *The Daily Dose: Early Usability of an LLM Tool for Patient Summaries and Trial Matching in Radiation Oncology*. Clinical and Translational Radiation Oncology, 2026.